How to read the tcpdump traffic log

This reading explains how to use tcpdump to analyze the network traffic related to the security incident.

```
14:18:32.192571 IP your.machine.52444 > dns.google.domain: 35084+ A?
yummyrecipesforme.com. (24)

14:18:32.204388 IP dns.google.domain > your.machine.52444: 35084
1/0/0 A 203.0.113.22 (40)
```

The first section of the DNS & HTTP traffic log file shows the source computer (**your.machine.52444**) using port **52444** to send a DNS resolution request to the DNS server (**dns.google.domain**) for the destination URL (**yummyrecipesforme.com**). Then the reply comes back from the DNS server to the source computer with the IP address of the destination URL **(203.0.113.22**).

```
14:18:36.786501 IP your.machine.36086 > yummyrecipesforme.com.http:
Flags [S], seq 2873951608, win 65495, options [mss 65495,sackOK,TS
val 3302576859 ecr 0,nop,wscale 7], length 0

14:18:36.786517 IP yummyrecipesforme.com.http > your.machine.36086:
Flags [S.], seq 3984334959, ack 2873951609, win 65483, options [mss
65495,sackOK,TS val 3302576859 ecr 3302576859,nop,wscale 7], length 0
```

The next section shows the source computer sending a connection request (**Flags [S]**) from the source computer (**your.machine.36086**) using port **36086** directly to the destination (**yummyrecipesforme.com.http**). The **.http** suffix is the port number; **http** is commonly associated with port 80. The reply shows the destination acknowledging it received the connection request (**Flags [S.]**). The communication between the source and the intended destination continues for about 2 minutes, according to the timestamps between this block (**14:18**) and the next DNS resolution request (see below for the **14:20** timestamp).

TCP Flag codes include:

Flags [S] - Connection **S**tart
Flags [F] - Connection **F**inish
Flags [P] - Data **P**ush
Flags [R] - Connection **R**eset
Flags [.] - Acknowledgment

```
14:18:36.786589 IP your.machine.36086 > yummyrecipesforme.com.http:
Flags [P.], seq 1:74, ack 1, win 512, options [nop,nop,TS val
3302576859 ecr 3302576859], length 73: HTTP: GET / HTTP/1.1
```

The log entry with the code **HTTP: GET / HTTP/1.1** shows the browser is requesting data from **yummyrecipesforme.com** with the **HTTP: GET** method using **HTTP** protocol version **1.1**. This could be the download request for the malicious file.

```
14:20:32.192571 IP your.machine.52444 > dns.google.domain: 21899+ A?
greatrecipesforme.com. (24)

14:20:32.204388 IP dns.google.domain > your.machine.52444: 21899
1/0/0 A 192.0.2.172 (40)

14:25:29.576493 IP your.machine.56378 > greatrecipesforme.com.http:
Flags [S], seq 1020702883, win 65495, options [mss 65495,sackOK,TS
val 3302989649 ecr 0,nop,wscale 7], length 0

14:25:29.576510 IP greatrecipesforme.com.http > your.machine.56378:
Flags [S.], seq 1993648018, ack 1020702884, win 65483, options [mss
65495,sackOK,TS val 3302989649 ecr 3302989649,nop,wscale 7], length 0
```

Then, a sudden change happens in the logs. The traffic is routed from the source computer to the DNS server again using port **.52444** (**your.machine.52444 > dns.google.domain**) to make another DNS resolution request. This time, the DNS server routes the traffic to a new IP address (**192.0.2.172)** and its associated URL (**greatrecipesforme.com.http**). The traffic changes to a route between the source computer and the spoofed website (outgoing traffic: **IP your.machine.56378 > greatrecipesforme.com.http** and incoming traffic: **greatrecipesforme.com.http > IP your.machine.56378**). Note that the port number (**.56378**) on the source computer has changed again when redirected to a new website.

Resources for more information

- An introduction to using tcpdump at the Linux command line: Lists several tcpdump commands with example output. The article describes the data in the output and explains why it is useful.

- tcpdump Cheat Sheet: Lists tcpdump commands, packet capturing options, output options, protocol codes, and filter options
- What is a computer port? | Ports in networking: Provides a short list of the most common ports for network traffic and their associated protocols. The article also provides information about ports in general and using firewalls to block ports.
- Service Name and Transport Protocol Port Number Registry: Provides a database of port numbers with their service names, transport protocols, and descriptions
- How to Capture and Analyze Network Traffic with tcpdump?: Provides several tcpdump commands with example output. Then, the article describes each data element in examples of tcpdump output.
- Masterclass – Tcpdump – Interpreting Output: Provides a color-coded reference guide to tcpdump output